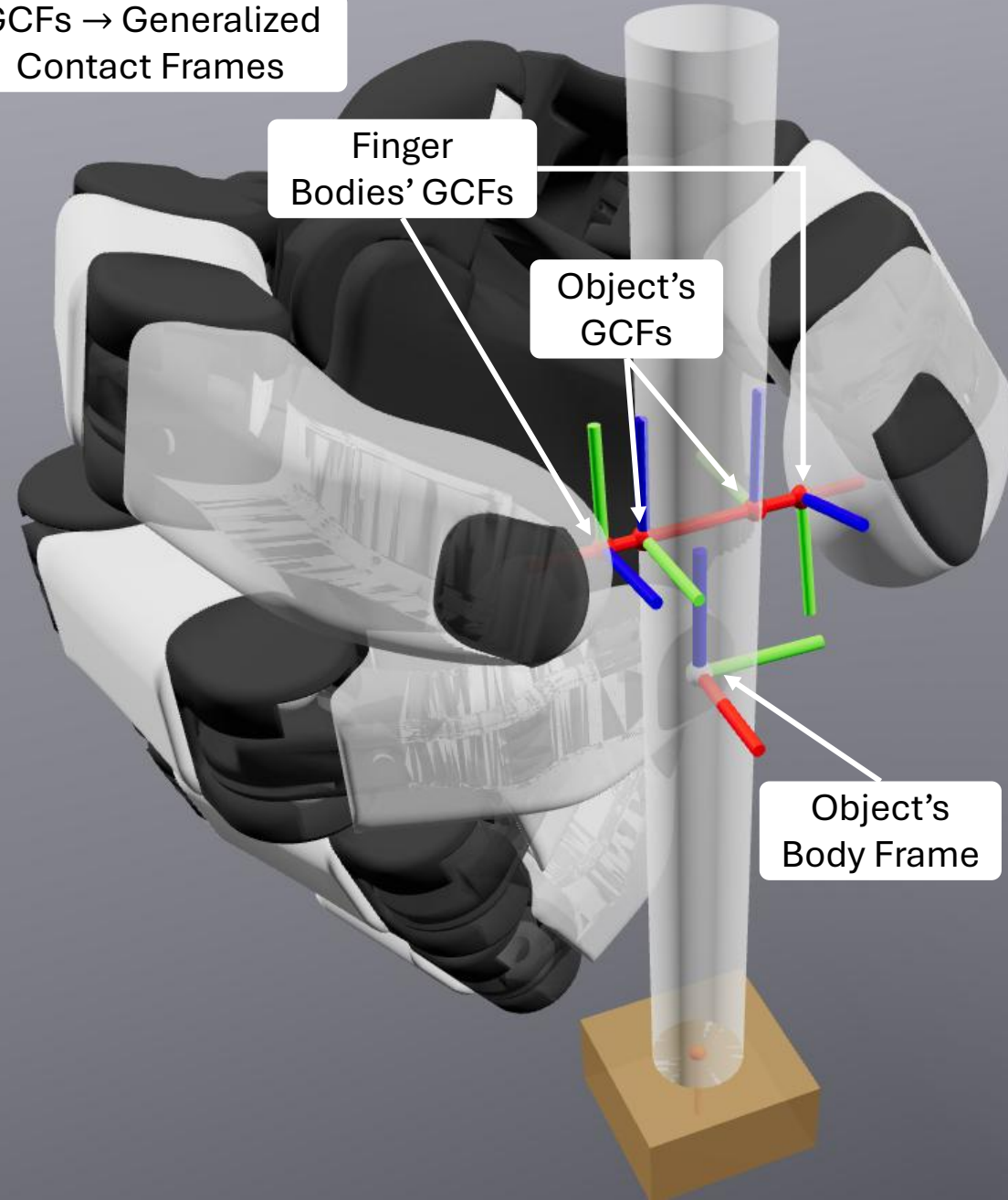


Hand-Object Mechanism = Hand + Object + GCFs

GCFs → Generalized Contact Frames



Palm wx [deg/s]		0
Palm wy [deg/s]		0
Palm wz [deg/s]		0
Palm vx [mm/s]		0
Palm vy [mm/s]		0
Palm vz [mm/s]		0
Palm Brake (B)		
Record Trajectories (R)		
Stop Simulation (ESC)		
Save Generalized Positions		
Realtime Rate		0.15
Object wx [deg/s]		0
Object wy [deg/s]		0
Object wz [deg/s]		0
Object vx [mm]		0
Object vy [mm]		0
Object vz [mm]		0
Object s		0
Sub-assembly 1		0
f1 [mm]		0
Finger 1 wx [deg/s]		0
Finger 1 wy [deg/s]		0
Finger 1 wz [deg/s]		0
Finger 1 vx [mm/s]		0
Finger 1 vy [mm/s]		0
Finger 1 vz [mm/s]		0
Sub-assembly 2		
f2 [mm]		0
Finger 2 wx [deg/s]		0
Finger 2 wy [deg/s]		0
Finger 2 wz [deg/s]		0
Finger 2 vx [mm/s]		0
Finger 2 vy [mm/s]		0
Finger 2 vz [mm/s]		0

Palm's reference angular velocity relative to world frame

Palm's reference linear velocity relative to world frame

$\omega_{PB_0}^b$, reference object body angular velocity relative to palm

$v_{PB_0}^b$, reference object body linear velocity relative to palm

s, object's reduced-order reference speed

Finger 1's sub-assembly selector (0 → pusher, 1 → pinch)

Finger 1's impedance controller's applied force magnitude

$v_{spin,ref}^{(1)}$, finger 1's reference spin velocity

$v_{roll,ref}^{(1)}$, finger 1's reference roll velocity

$v_{sep,ref}^{(1)}$, finger 1's reference separation velocity

$v_{slide,ref}^{(1)}$, finger 1's reference slide velocity

Finger 2's

- sub-assembly selector
- impedance controller's applied force magnitude
- 6 scalars in reference relative contact angular and linear velocities

Teleoperation Interface

Form reference object body twist,

$$V_{PB_0}^b = \begin{bmatrix} \omega_{PB_0}^b \\ v_{PB_0}^b \end{bmatrix}$$

Form finger 1's reference relative contact angular velocity,

$$\omega_{L_0L_1,ref}^{b(1)} = \begin{bmatrix} v_{spin,ref}^{(1)T} & v_{roll,ref}^{(1)T} \end{bmatrix}^T$$

Form finger 1's reference relative contact linear velocity,

$$v_{L_0L_1,ref}^{b(1)} = \begin{bmatrix} v_{sep,ref}^{(1)T} & v_{slide,ref}^{(1)T} \end{bmatrix}^T$$